Hrishikesh Anand

Computer Science Undergraduate | IoT • Machine Learning • Full-Stack Development

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SUMMARY

Computer Science undergraduate at VIT Vellore who builds end-to-end systems, from firmware on ESP32 microcontrollers to machine learning models and cloud-deployed web apps. Five live projects across IoT, ML and full-stack development. Seeking an entry-level software engineering role.

EDUCATION

Vellore Institute of Technology (VIT), Vellore **2023 – 2027**
B.Tech. in Computer Science and Engineering Vellore, Tamil Nadu

TECHNICAL SKILLS

Languages	Java, Python, C/C++, JavaScript, TypeScript, SQL
Backend & Databases	FastAPI, Node.js, Express, PostgreSQL, Firebase, Firestore
Machine Learning	scikit-learn, Pandas, NumPy, SHAP, Streamlit, Matplotlib
Frontend	React, Tailwind CSS, Chart.js
Tools & Deployment	Git, GitHub, Vercel, Render
Core CS	Data Structures & Algorithms, Object-Oriented Programming
Embedded & IoT	ESP32, Arduino, FreeRTOS, I2C sensors

PROJECTS

Event Data Recorder for Light Electric Vehicles GitHub | Live Demo

ESP32, C++, Firebase, JavaScript, Chart.js, Vercel

- Engineered an ESP32-based “black box” for e-scooters and e-bikes that detects battery overheating, gas leaks, impacts and free-falls in real time.
- Designed a priority-based event-logging pipeline that pushes critical incidents to Firebase first and visualises them on a live Chart.js dashboard.
- Deployed the monitoring dashboard on Vercel for remote, real-time access to vehicle events.

Smart Health Band – Secure, AI-Assisted Vital-Sign Monitor GitHub | Live Demo

ESP32, C++, Python, scikit-learn, FastAPI, PostgreSQL, Firebase

- Co-developed a team wristband prototype that scores health risk on-device against each wearer’s personal baseline, using a compact ML model sized for the ESP32.
- Secured the data path with HMAC-SHA-256 signed readings, verified by a FastAPI backend before storage in PostgreSQL and Firebase.

FinAI 2.0 – AI-Driven Personal Finance Manager GitHub | Live Demo

React, TypeScript, Node.js/Express, FastAPI, Chronos-2, Gemini, Firestore, Render

- Built a full-stack finance app that forecasts spending with the Chronos-2 time-series model and computes the lowest-effort path to a savings goal.
- Added Gemini-generated explanations guarded by a grounding check that blocks hallucinated numbers.

Heart Disease Prediction System GitHub | Live Demo

Python, scikit-learn, SHAP, Streamlit, Pandas, Matplotlib

- Trained a classifier on the UCI heart-disease dataset and deployed it as a Streamlit app, with SHAP plots showing which clinical factors drive each prediction.

DSA Visualizer GitHub | Live Demo

Java, React

- Built an interactive web tool that animates sorting and searching algorithms step by step, so learners can follow exactly how each one works.